

Revon-H System Architecture

How the frontend, local API, content-addressed core, SQLite persistence, and reproducible evidence pipeline fit together.

End-to-end platform



Component	Primary responsibility	Implementation
Frontend	Collect actions and visualize history, differences, work, storage, and integrity.	React / Vinext and a typed fetch client.
HTTP API	Validate requests, constrain repository names, map errors, and serialize writes.	Python standard-library HTTP server.
Revon-H	Create versions, incrementally copy trie paths, check out history, and compare versions.	VersionedDatabase.
Persistence	Store immutable objects and mutable references in one durable transaction.	SQLiteRevonRepository.
Evaluation	Execute fixed-seed workloads over baselines, Revon variants, and Dolt.	experiments/final_benchmark.py.

Trust and deployment boundary

The complete interactive demo is intentionally local-first. The backend binds to 127.0.0.1, validates repository names, verifies stored hashes, and accepts configured browser origins. Authentication, TLS, rate limiting, and multi-host coordination are outside the current prototype.

Research contribution boundary

The research contribution is the inspectable combination of a fixed-depth Merkle hash trie, incremental copy-on-write updates, addressed changesets, and adaptive Log/Merkle selection. SQLite supplies durability; the frontend supplies interaction; the benchmark harness supplies comparative evidence.

Object model and commit path

Immutable identities are separated from mutable aliases and operational measurements.

Addressed graph

Object or reference	Contains or points to	Identity behavior
Commit	Root hash, parent commit, changeset hash, message, timestamp.	Canonical payload determines commit hash.
Trie root/internal node	Numbered child slots containing child hashes.	Canonical child map determines node hash.
Leaf	Canonical key/value entries for one route bucket.	Canonical entries determine leaf hash.
Changeset	Addressed put/delete records for one commit.	Canonical operations determine changeset hash.
Version alias	Monotonic display number mapped to a commit hash.	Human-readable alias; not content identity.
HEAD	Latest commit hash.	Mutable reference advanced transactionally.

Commit request



A failed persistence statement causes a complete SQLite rollback. The wrapper then reloads the last durable state, preventing the in-memory HEAD from moving ahead of the repository file. Reopen verification rejects corrupted payloads or broken references.

Read requests

Request	Core path	Result
Checkout	Resolve version to commit, then materialize the addressed root.	Historical key/value state plus pagination metadata.
Compare	Resolve endpoints, then run forced Log, forced Merkle, or Hybrid.	Added, modified, and deleted keys plus natural work metrics.
Metrics	Read commit and repository statistics; optionally run a version comparison.	Storage, sharing, write, and diff measurements.
Open	Load metadata and objects, verify hashes and graph invariants.	HEAD, integrity counts, and a usable repository handle.

Hybrid diff and evidence flow

Runtime defaults and paper parameters are intentionally distinguished.

Hybrid decision



Configuration	Value	Scope
Runtime Hybrid threshold	128 operations	Default in the core, durable store, API, and CLI demo.
Paper Hybrid threshold	4,096 operations	Frozen calibrated boundary for the audited paper machine and workload family.
Trie geometry	b=8, d=4	Runtime and paper default; evaluated against four other configurations.

Evidence pipeline



The versioned final evidence bundle contains 333 successful and correct measured rows. The separate sensitivity bundle contains 45 correct trials across five geometries. Paper tables and figures are generated from reviewed summary CSV files and checked against raw results and manifests.

Local run sequence

Start `python -m revon_api`, then run `npm run dev` from `frontend/`. For automated verification, run `python -m unittest -v` and `npm test`.

Repository

Source, evidence, and reproduction instructions: <https://github.com/atharvasheersh/Revon>